

Pokémon Red benchmark: Jev alone vs Jev + GPT-6 Sol

28 September 2026 · runs tagged **v2** · sampling temperature **0.5**

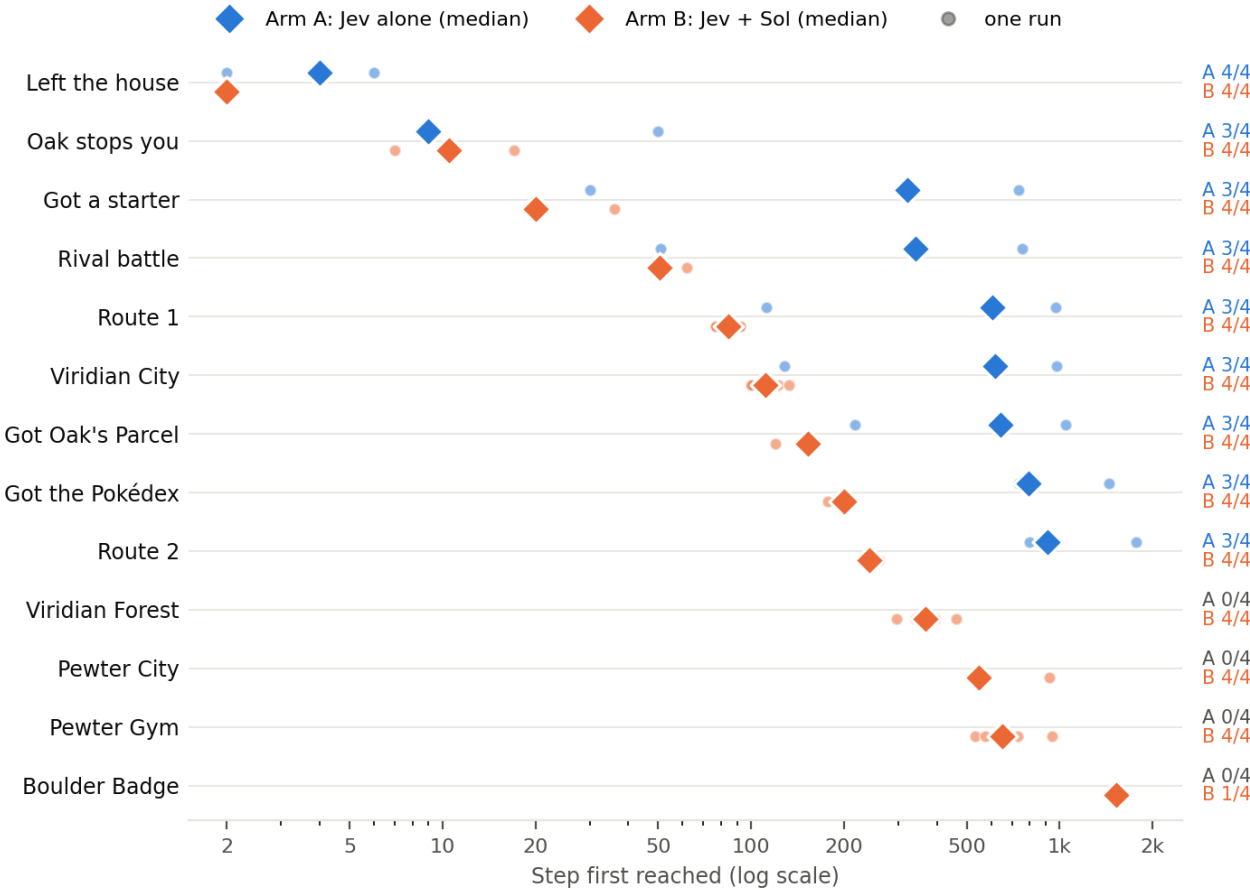
Both arms play from the same new-game save with the same harness. Jev (TypeSafe, via OpenRouter) makes every decision in both arms. In Arm A its goal is fixed (“make progress in the main story toward the next badge”); in Arm B, GPT-6 Sol reads the game state every 15 steps and rewrites that goal. Each run is capped at 2,000 steps and ends early after 1,000 steps without a new milestone.

	Arm A: Jev alone	Arm B: Jev + Sol
Average milestones (of 13)	7.00	12.25
Runs that beat Brock (badge 1)	0 of 4	1 of 4
Average blackouts per run	2.8	9.2
Average cost per run	\$0.09	\$0.62

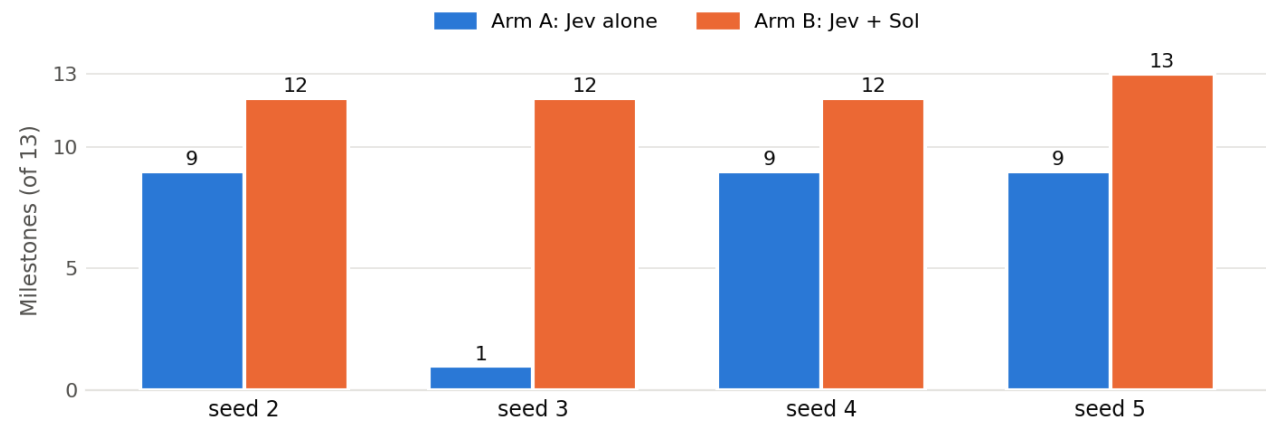
Seeds 2, 3, 4, 5 (4 runs per arm).

How fast each arm reached each milestone

Each small dot is one run; the diamond is the median. The column on the right counts how many runs reached that milestone at all. The step axis is logarithmic so the early milestones stay readable.



Milestones reached per run



Median step per milestone

Median over the runs that reached it; brackets show how many runs did. Speed-up = Arm A median ÷ Arm B median.

Milestone	Arm A median step	Arm B median step	Speed-up
Left the house	4 (4/4)	2 (4/4)	2.0×
Oak stops you	9 (3/4)	10.5 (4/4)	0.9×
Got a starter	320 (3/4)	20 (4/4)	16.0×
Rival battle	341 (3/4)	50.5 (4/4)	6.8×
Route 1	606 (3/4)	84.5 (4/4)	7.2×
Viridian City	616 (3/4)	111.5 (4/4)	5.5×
Got Oak's Parcel	642 (3/4)	152.5 (4/4)	4.2×
Got the Pokédex	791 (3/4)	200.5 (4/4)	3.9×
Route 2	912 (3/4)	241.5 (4/4)	3.8×
Viridian Forest	- (0/4)	368 (4/4)	-
Pewter City	- (0/4)	548.5 (4/4)	-
Pewter Gym	- (0/4)	652 (4/4)	-
Boulder Badge	- (0/4)	1527 (1/4)	-

Every run

Arm	Seed	Steps	Score	Starter	Rival	Blackouts	Ember uses	Cost
A	1*	2000	10	Charmander	lost	4	0	\$0.111
A	2	1798	9	Charmander	won	7	0	\$0.100
A	3	1006	1	-	-	0	0	\$0.056
A	4	2000	9	Squirtle	won	2	0	\$0.108
A	5	1912	9	Charmander	won	2	0	\$0.109
B	1*	2000	12	Charmander	won	1	3	\$0.771
B	2	1732	12	Charmander	won	7	15	\$0.592
B	3	1533	12	Charmander	lost	13	9	\$0.542
B	4	1947	12	Charmander	won	10	12	\$0.649
B	5	2000	13	Charmander	won	7	37	\$0.685

* Seed 1 ran before a map-name fix: its “Viridian Forest” milestone was really the forest gate, and Arm B’s “Pewter Gym” was really the museum’s second floor. It is excluded from the charts and averages above.

Does the temperature change the result?

The same seeds run at two sampling temperatures, $T = 0.5$ and $T = 1$. Median steps are over the runs that reached each milestone (brackets: how many did).

	Arm A, $T=0.5$	Arm A, $T=1$	Arm B, $T=0.5$	Arm B, $T=1$
Average milestones	7.00	8.75	12.25	12.75
Beat Brock	0 of 4	0 of 4	1 of 4	3 of 4
Left the house	4 (4/4)	3.5 (4/4)	2 (4/4)	2.5 (4/4)
Oak stops you	9 (3/4)	33.5 (4/4)	10.5 (4/4)	16.5 (4/4)
Got a starter	320 (3/4)	49.5 (4/4)	20 (4/4)	35.5 (4/4)
Rival battle	341 (3/4)	75 (4/4)	50.5 (4/4)	64.5 (4/4)
Route 1	606 (3/4)	119.5 (4/4)	84.5 (4/4)	106 (4/4)
Viridian City	616 (3/4)	149 (4/4)	111.5 (4/4)	139 (4/4)
Got Oak's Parcel	642 (3/4)	496 (4/4)	152.5 (4/4)	160 (4/4)
Got the Pokédex	791 (3/4)	602 (3/4)	200.5 (4/4)	207 (4/4)
Route 2	912 (3/4)	688 (3/4)	241.5 (4/4)	325 (4/4)
Viridian Forest	- (0/4)	1564 (1/4)	368 (4/4)	451 (4/4)
Pewter City	- (0/4)	- (0/4)	548.5 (4/4)	733 (4/4)
Pewter Gym	- (0/4)	- (0/4)	652 (4/4)	738 (4/4)
Boulder Badge	- (0/4)	- (0/4)	1527 (1/4)	1507 (3/4)
Average cost per run	\$0.09	\$0.08	\$0.62	\$0.61

What the results say

The planner roughly doubles how far Jev gets, and gets there several times faster. Across seeds 2–5, Arm B averaged 12.25 of 13 milestones against 7.0 for Arm A. Every Arm B run reached Viridian Forest, Pewter City and the Pewter Gym; no Arm A run got past Route 2. One Arm B run (seed 5) beat Brock and earned the Boulder Badge at step 1,527.

The biggest gap is the first bit of "quest logic": choosing a starter. Once Oak asks which Pokémon you want, Arm B picks one at a median of step 20; Arm A takes a median of 320 steps (16× slower), spending hundreds of steps pressing A at Oak, and one run (seed 3) never got there at all — it wandered into Oak's Lab before triggering Oak and then talked to Blue about 800 times. Later milestones are still reached 4–7× faster by Arm B. (Update: most of this starter gap turned out to be caused by the sampling temperature; see "Update: the same test at temperature 1" below.)

Arm A is not stuck forever — it is slow. Three of four Arm A runs eventually chose a starter by chance (sampling occasionally sends Jev to a Poké Ball) and then completed Oak's Parcel errand and got the Pokédex on their own. Arm A's limit in 2,000 steps was Route 2.

Behaviour worth noting

Jev fights on autopilot. In both arms Jev almost always presses A in battle, which picks FIGHT and then the first move. Arm A never used Ember in any run, even after learning it; Arm B used it 9–37 times per run, because Sol's goals sometimes mention it. Charmander's Ember is what beat Brock's rock types in seed 5.

Blackouts are higher in Arm B, but that reflects how far it got. Arm B blacked out 7–13 times per run versus 2–7 for Arm A, because it spent its time in Viridian Forest and the Pewter Gym, where trainers are much tougher than Route 1 wild Pokémon. Neither arm heals deliberately: Jev keeps fighting on low HP until it faints, and a blackout sends it back to a Pokémon Center.

Where Arm B stalls: the Pewter Gym. All four clean Arm B runs entered the gym, but three of them were beaten repeatedly by the gym trainer or Brock and never got the badge within the step budget. Winning needs levelling up and using Ember, which the planner only partly achieves.

Cost. Arm A cost about \$0.09 per run; Arm B about \$0.62, of which about \$0.53 is Sol. Per milestone reached, that is about \$0.013 for Arm A and \$0.050 for Arm B.

Update: the same test at temperature 1

After this report was first written, both arms were re-run on the same seeds at temperature 1, where options are picked at exactly Jev's own probabilities instead of being sharpened towards its favourite. The full comparison table is on the page just before these findings.

The planner's advantage holds. At temperature 1, Arm B averaged 12.75 milestones against 8.75 for Arm A, and beat Brock in 3 of 4 runs (up from 1 of 4). No Arm A run reached Pewter City at either temperature.

But the starter gap was mostly caused by the temperature. At temperature 1 Arm A chose a starter at a median of step 49.5 against 35.5 for Arm B — only 1.4× slower, not 16×. Sharpening had squashed Jev's small chance of walking to a Poké Ball, trapping Arm A in the "talk to Oak" loop. Later milestones still favour Arm B by 2–3× (Oak's Parcel 3.1×, Pokédex 2.9×, Route 2 2.1×).

Temperature 1 was better for both arms (Arm A 7.00 → 8.75, Arm B 12.25 → 12.75), so it is the better default for future runs, and any "N× faster" figure should be quoted with its temperature.

Caveats

Four clean runs per arm is a small sample, so treat the averages as indicative rather than precise. Runs stop after 1,000 steps without a new milestone, so a slow arm is cut off earlier than it might otherwise be. Button choices are sampled from Jev's probabilities, which is why identical settings give different runs; as the temperature-1 update shows, the temperature changes Arm A's early-game numbers substantially. Seed 1 is excluded because it ran before a map-name fix.

Suggested next steps

Re-run with more seeds (e.g. 10 per arm) at temperature 1 on harness v3 to tighten the averages. Give Arm A a much larger step budget (it is cheap at about \$0.05 per 1,000 steps) to test whether it eventually reaches Pewter. Try re-planning when Jev's confidence stays low, which the spec suggests, to see whether Arm B can get through the gym more reliably.